

- “Tell me the name of student at index 7.”
- “Find the student whose card equals 7.”
- “Insert a new student at position 0.”
- “Swap students at positions 2 and 8.”
- Read $a[k]$ in an array
- Find max in an array
- Swap $a[i]$ and $a[j]$
- Insert at end of an array when there’s free capacity
- Insert at middle of an array
- Remove last element of array
- Remove first element of array
- Compute sum of first n numbers with a loop
- Compute sum of first n numbers with formula $n * (n+1)/2$

- If we know the index?
- If we don’t know the index and must search
- If we must touch every element once
- If we add only a constant number of variables